

Transcripts: First evaluation round

Expert 1 at Hawk

Date: Auguste 9th, 2024

Time: 14:15

Place: Microsoft Teams

00:00:00 – 00:24:57

Interviewer Perfect. So, next, I would like to go through my design principles with you. As I mentioned earlier, I'd appreciate it if you could be really critical and provide feedback. Some of the principles might be a bit complex or formal in their wording, but that's because they are based on how design principles are typically presented in the literature. Even if they seem complicated, please feel free to share your thoughts – whether it's about the content, correctness, or language. I'm open to all your feedback. Let's start with the first design principle, which is focused on adaptive learning and continuous improvement. The idea here is to enhance fraud detection accuracy and performance over time, particularly for compliance officers dealing with evolving transactional behaviors and changing card user habits. This principle emphasizes the importance of employing real-time learning mechanisms to analyze historical data and continuously integrate user feedback. The goal is to implement a constant updating and refinement process that dynamically adapts to both emerging fraud patterns and normal variations in card user behaviors. This approach ensures that the fraud detection system remains accurately tuned to both fraudulent activities and legitimate user trends, thereby enhancing the system's reliability in our rapidly changing environment. So, my question to you is: What do you think about this first design principle? #00:00:04#

Interviewee I guess when you mention real-time learning mechanisms, you're referring to something that updates automatically, like an online learning system, right? That's something we consciously decided not to implement due to compliance-related reasons. You can lose control over the process, and there's no guarantee that the model will develop in the desired direction. Without human oversight, the model could change in an unfavorable way, and it might be too late before anyone notices. That's why we continuously retrain the models at regular intervals. However, the final decision always rests with the customer, they decide whether or not to apply the new model. This way, they remain in full control. It's not like something bad happens due to the online learning system not working properly, or because poor labels were provided by an analyst, leading the model to deteriorate without anyone noticing. #00:02:03#

Interviewer So instead of saying real-time learning mechanism, you could say that it's important to always use the latest data available to train the model. However, this doesn't mean deploying it in real time. It's more about updating and adapting the model. Ultimately, the decision to deploy and use the updated model is not done in real time, it's still up to the client to decide if they want to implement the newest version. #00:03:16#

Interviewee Yes. #00:03:32#

Interviewer The next point is the desired result number two: user-centric development and alignment. This principle is about creating a tailored fraud detection system that aligns with a business's model, goals, services, and customer needs, whether it's for managers, compliance officers, or end users like us. In the context of fraud detection system development, it's crucial to actively involve end users and establish regular feedback loops. This approach allows the system to be customized to specific business needs, ensuring that the fraud detection system is effectively aligned with strategic objectives and optimizes the cost-benefit ratio for each end user. This can include meeting specific goals such as reducing manual work, increasing efficiency or speed, enhancing risk management, or making optimal use of available resources. This is particularly important for businesses with limited human resources, where it's not feasible to generate as many alerts or handle as many cases as larger organizations, like Citigroup, which has entire departments dedicated to fraud detection. As we've discussed many times, every business has different needs, so there isn't a one-size-fits-all fraud detection system. Each system needs to be tailored to the specific requirements of the business. What do you think about this design principle? #00:03:40#

Interviewee I fully agree. This is something I always emphasize as well. When we develop models, we should use metrics that align with the business goals we want to achieve. It doesn't make sense to optimize a metric

that measures the accuracy of the model if it's not directly related to the specific business objective. So, I completely agree with that point. #00:05:03#

Interviewer Thank you. The next principle focuses on the development of comprehensive data integration and quality assurance. This is particularly important for enhancing fraud detection accuracy for compliance officers by ensuring data quality and integrity. It's crucial to have a full understanding of the data, employ comprehensive preprocessing techniques, apply advanced feature engineering, and ensure secure data handling and integration. Additionally, it's important to analyze various contextual data points, not just the transaction itself, but also factors like time, location, logging behaviors, and device information. By considering the full context of each transaction, this approach ensures that the model is built on a robust and reliable data foundation, allowing for more accurate identification of fraudulent activities. What do you think about this design principle? #00:05:34#

Interviewee I mean, again, this sounds very good. Just thinking if there's anything that I wouldn't agree with. I think, no, I think this makes sense. Because data quality is the starting point for everything. #00:06:26#

Interviewer It's really about understanding the data. Before you can even develop a model, you need to fully understand all the data you have, including the data provided to you, before you can proceed. #00:06:50#

Interviewee Exactly. Maybe the only thing I'd point out about this design principle is that it seems to mix a few different concepts. On one hand, it talks about data understanding, quality, and integrity, but then it also mentions secure data handling, which I think is a different topic. For example, feature engineering is more closely related to data understanding. I'm not sure, while I fully agree with everything that's written, I might consider sharpening the message a bit and focusing it more. Secure data handling, for instance, sounds more like an information security issue, which might be slightly off-topic here. #00:07:00#

Interviewer Right, it's about using the information provided by all the clients while ensuring that the data used to train the model is handled properly. I assume that when you train the model, the data is mostly anonymized or restricted in some way. As a developer, you need to ensure confidentiality, it's about not exposing sensitive data. In the end, it's crucial that the data from our clients is protected. #00:07:57#

Interviewee Well, exactly, but that feels like a slightly different topic to me. So, one topic is really about understanding the data and deriving the right signals from it, while the other topic is more of an information security issue. For me, these are almost two different design principles, not a single one. #00:08:23#

Interviewer That makes sense. I understand what you mean. The other one is more about the process, like preprocessing, training, and all that. It really should be another design principle if you want to focus on data security in the end. Perfect. The next one is the hybrid detection approach. This principle is about enhancing detection efficiency and adaptability for compliance officers by employing a hybrid approach that integrates AI and rule-based methods. The idea is that this combination provides both flexibility in adjusting detection rules and precision in fraud detection, leveraging the strengths of both methodologies to improve accuracy and reduce the manual review workload. What I mean is that it doesn't make sense to rely solely on a rule-based approach or just an AI-based approach; it's more effective to use a combination of both. This ensures the system remains effective and responsive, allowing it to adapt to changing requirements over time. What do you think about this design principle? #00:08:56#

Interviewee I think there are some signals that are best addressed with rules, particularly for very black-and-white situations. For example, if there are regulatory requirements that mandate checking whether a transaction is from a high-risk country, that's obviously best handled with rules. But apart from these very specific use cases, I would personally lean towards an AI-only approach. I've seen this work effectively in some large-scale projects for which we've developed models, and from my perspective, AI is strictly superior, except in those special circumstances where you have a clear-cut scenario. However, I also think you can't completely eliminate rules because if you're introducing a completely new risk scenario for which you don't have any labelled data, you need a rule to bootstrap the process. Initially, you would use a rule to collect labels, and then you can train an AI model on that data. So, in that sense, you're right, rules will always have a place, but for me, that place is smaller than what the design principle suggests. The two main reasons I see for using rules are, first, to handle black-and-white cases, and second, to onboard new risk scenarios where labelled data isn't yet available. That said, most of our customers still prefer having rules, which is why we include them. But personally, I think AI is the superior approach, so I'd favor a heavier emphasis on AI rather than a 50/50 split or anything like that. #00:10:03#

Interviewer But do you think with a hybrid approach, in the future, it will be completely AI, or will there still be, say, 10% of rules in place? Do you think we'll ever reach a point where we can rely solely on AI, or will there always be a need for some rules? Maybe it could be better formulated, not as a 50/50 split, but more like having a small percentage of rules in place while AI takes over the majority of the work. #00:12:14#

Interviewee From my perspective, as I mentioned, there are those black-and-white scenarios where rules make sense, especially if you want to bootstrap the AI. For example, suppose you've never had a pass-through rule in your system, and you've trained a supervised AI model on operator labels for identifying false positives

or other scenarios. Since you've never had a pass-through rule, the model wouldn't understand what a pass-through is. In that case, you first need to implement a pass-through rule to collect the necessary labels, and then you can train the model using those labels. Once the model is trained, you can discard the pass-through rule because its purpose was just to bootstrap the AI. This allows the AI to handle those situations in the future without the need for the rule. So, for me, these are the two legitimate reasons why we'll always have some rules in place, even 10 years from now. I don't think rules will completely go away, but their role will be much more limited compared to what most of our customers are doing at the moment. However, I've seen a few AI approaches that completely outperform traditional rules-based systems, which is why I really believe the future is almost purely AI-driven. That's just my opinion, at least. #00:12:45#

Interviewer And maybe I could work on the formulation here to make the point clearer. But thank you so much. The next design principle is scalability and flexibility. The idea here is to maintain long-term usability and effectiveness for compliance officers, particularly in the context of increasing transaction volumes, which lead to growing data volumes, and the evolving nature of fraud techniques. This principle emphasizes the need to employ scalable infrastructure, modular system design, and regular performance assessments. The goal is to ensure that the fraud detection system can efficiently handle growth, adapt to new fraud methodologies, and remain effective over time. What do you think about this design principle? #00:14:14#

Interviewee I got nothing to add. It makes a lot of sense. You could maybe also include something about latency. I mean, it goes in a similar direction because for fraud detection, it's not only important to process large volumes but also to process each transaction quickly enough to potentially block it in real time. So, I would suggest adding the latency angle as well, but other than that, I'm super fine with it. #00:15:01#

Interviewer The next one is design principle six, which focuses on dynamic and interactive explainability. This principle aims to enable dynamic exploration of data and provide interactive explainability features for compliance officers using the fraud detection system. The idea is to offer user-friendly, interpretable insights, clear reasoning paths, and confidence levels in the system's decisions. This approach enhances trust, allows users to understand and validate the system's outputs, and supports effective decision-making when handling fraud cases. What do you think about this design principle? #00:15:29#

Interviewee I fully agree with that. That's definitely a cornerstone for sure. #00:16:16#

Interviewer Great, thank you. The next design principle, number seven, is about integrating preventive and detective measures. What I mean by this is enabling a strategic shift to combining both preventive and detective fraud management within our comprehensive fraud protection framework. The idea is to employ fraud detection system algorithms that support both preventive and detective measures. This approach allows the system to use the same well-trained model, ensuring more secure protection for card users and the company. In the long term, it doesn't make sense to rely solely on a detective approach. Once we have enough data and well-trained models, it's important to integrate a preventive approach as well. This will provide better protection for us and, of course, for our card users. What do you think about this design principle? #00:16:26#

Interviewee I need to understand a bit better how you draw the line between preventive and detective measures. Could you maybe elaborate on that? #00:17:24#

Interviewer Preventive measures are like real-time detection, where we block a transaction as soon as we get the request by analyzing it immediately and deciding if it's fraud or not. If it's identified as fraud, the transaction is blocked on the spot. Detective measures, on the other hand, involve accepting the transaction first, then applying rules to either block or approve it afterward. Currently, at amnis, this is the approach we use, every transaction is analyzed post-acceptance to determine if it's fraudulent. In the long term, like the big companies, Visa, MasterCard, or Citibank, they also implement preventive measures, blocking transactions before they're even accepted. So, it makes sense to have a strategic shift towards incorporating both approaches to ensure better security and protection for our customers. #00:17:33#

Interviewee Ok, then. I fully agree with that. I fully agree with that. #00:18:29#

Interviewer Maybe I need to refine it a bit more. I'm not entirely happy with that formulation either, but the key point is about having both approaches in place in the end, you know? #00:18:33#

Interviewee No, it makes sense. I just wasn't sure what you meant by preventive. Maybe you could mention that by preventive, you mean blocking transactions in real-time, right? That's what you're referring to, exactly. And I think for fraud, that's, of course, super, super important for the reasons you mentioned. So, I agree. #00:18:43#

Interviewer The last design principle is about bias mitigation and ethical decision-making. To ensure fair treatment of all card users and minimize human bias in the decision-making process, it's important to employ

algorithms that detect and measure bias, perform continuous monitoring, and conduct regular audits and updates. This approach ensures that decisions are made without undue bias, fostering trust and reliability in the fraud detection system. We're addressing bias in both the training data and in human decision-making for example, if a compliance officer repeatedly marks obviously fraudulent transactions as not fraudulent. We aim to mitigate bias from both the human side and the training data. What do you think about this design principle? #00:19:04#

Interviewee Yes, that makes sense. This is a topic we've discussed before, and I personally try to see it from a nuanced perspective. I think it also depends on the type of model you're building. For instance, if a false positive reduction model uses data fields such as gender or nationality, I see it as less of an issue compared to a model that uses these fields for opening accounts. In a false positive reduction model, the impact is minimal, the worst that happens is someone might not get auto-closed, while another person might. If the one not getting auto-closed is from a high-risk country, it doesn't have a direct effect on their life because a human will still investigate the transaction afterward. So, nothing significant has occurred yet. For false positive reduction models, I'm more open to using data fields that could potentially introduce bias. It may sound bad to introduce bias, but sometimes these fields add value and improve the model's performance, even if they could lead to discrimination in some way. For example, if data shows that people from a specific city, like Berlin, are statistically more likely to engage in financial crime than those from Hamburg, that's just a fact. However, I would differentiate this from models used for account openings, which require higher standards. For opening models, especially preventive ones that block transactions, introducing bias is a much bigger issue. If such a model discriminates, it could lead to real discrimination in someone's daily life, like consistently blocking their transactions. In these cases, the highest standards regarding discrimination must be applied. So, my only comment would be that while I allow the use of such data fields in false positive reduction models, I always tell my team to ensure that the model isn't making decisions solely based on someone's origin, gender, or similar factors, it should just be one factor among many. But for opening models, we usually don't use these discriminatory fields at all and focus solely on transaction behavior. Does that make sense? #00:20:07#

Interviewer Perfect. So, in general, do you think all the critical aspects of developing a fraud detection system are covered with these design principles? Also, do you have any general feedback about these design principles? #00:23:17#

Interviewee Maybe there's another aspect to consider, the transparency during the model training process, which ties into the explainability topic. It's important to make the entire process as transparent as possible, meaning clear model documentation. In the end, it's about detailing what features the model is trained on, how the model was validated, how the back tests were conducted, and so on. I think a good fraud detection system should make all of these steps fully transparent. #00:23:36#

Interviewer We have explainability, but transparency is missing. That's a good point. #00:24:06#

Interviewee Transparency during model training is important because explainability makes the inference process transparent. #00:24:19#

Interviewer That's good. #00:24:20#

Interviewee When the model is executed, you also want to make it transparent how the model was trained to begin with. #00:24:27#

Interviewer OK, sounds good. This was definitely a topic that was addressed everywhere in the literature, but somehow it got overlooked. That's the reason why getting this feedback is so valuable, you know. I can't think of any other points that were missed. Perfect. Thank you so much. We're done. #00:24:33#

Interviewee Thank you, too. #00:24:55#

Expert 2 at Hawk

Date: Auguste 7th, 2024

Time: 15:15

Place: Microsoft Teams

00:00:00 – 00:26:22

Interviewer As I mentioned, let's start with the first design principle. This one is about adaptive learning and continuous improvements. To improve fraud detection accuracy and performance over time for compliance officers and end users, particularly in the context of evolving transactional behaviors and changing card user habits, it's important to employ real-time learning mechanisms, analyze historical data, and continuously integrate user feedback. For example, feedback from compliance officers indicating whether a case was fraudulent or not. This involves implementing a constant updating and refinement process that dynamically adapts to both emerging fraud patterns and normal variations in card user behaviors. This approach ensures that the fraud detection system remains accurately tuned to both fraudulent activities and legitimate user trends, enhancing the system's reliability in a rapidly changing environment. I know the wording is highly complicated, but this is the required format for how they need to be formulated. I'd like to know what you think about this first design principle. #00:00:04#

Interviewee For adaptive learning and continuous improvement, that makes perfect sense to me. I don't see any reason why this wouldn't be beneficial. Having a constant update and refinement process is important, and maybe it would be useful to tie it to a specific metric, right? Because I guess banks and financial institutions usually work in a number-based space. Maybe we could add something about continuous monitoring of certain metrics, like fraud rates or true positive rates. For instance, you could set up a process within the organizational structure where these rates, like fraud rates, true positive, or false positive rates, are regularly monitored. This way, we can see if we are improving detection or if it's declining, possibly due to new fraud trends. If the fraud detection models are supervised, their performance might decrease over time as new fraud patterns emerge. So, I would suggest adding some continuous monitoring, like establishing a process for metric-based monitoring, really focusing on the numbers. #00:01:19#

Interviewer Perfect. Thank you so much. The next design principle is user-centric development and alignment. It's important to create a tailored fraud detection system that aligns with the business model, goals, services, and customer base for end users like us. In the context of the detection system development process, it's crucial to actively involve end users and establish regular feedback loops, so the system is tailored to specific business needs. This ensures that the fraud detection system is effectively aligned with strategic objectives and optimizes the cost-benefit ratio for each end user. As I mentioned, this can be achieved by meeting specific goals such as reducing manual work, increasing efficiency or speed, enhancing risk management, and making optimal use of available resources. What do you think about this design principle? #00:02:27#

Interviewee I really like the concept of regularly having touchpoints with the users. This approach doesn't just apply to fraud detection systems but to any kind of software development where there's direct user interaction. Maybe we could emphasize this a bit more. I think you mentioned it somehow, but I'll put it in my own words, it's important to set up regular touchpoints with the actual end users, not just the managers. By end users, I mean the people who work directly with the system. We should find out what else they are doing that our system isn't currently offering. For example, are they looking up transaction history in another database because some data points are missing? Or are they resorting to Google searches? If that's the case, maybe we could integrate such features into the application, so they don't have to go outside the system. This could even help improve fraud assessment by leveraging that information directly within the system. So, in summary, regular touchpoints with the end users, not just the managers, are crucial for understanding how they interact with the system and identifying potential areas for improvement. #00:03:30#

Interviewer Thank you so much again. The next design principle is comprehensive data integration and quality assurance, particularly focusing on development. To enhance fraud detection accuracy for end users, especially in the context of data quality and integrity, it's important to employ thorough data understanding, comprehensive preprocessing techniques, advanced feature engineering, secure data handling, and integration of various contextual data points. This means not only analyzing the transaction data itself but also considering all relevant contextual data, such as time, location, logging behaviors, device ID, or IP address. It's crucial to incorporate these contextual elements because they ensure that

the model is built on a robust and reliable data foundation, which is essential for accurately identifying fraudulent activities. What do you think about this design principle? #00:05:02#

Interviewee Absolutely, this is hugely important. You wouldn't believe how many issues arise, or aren't even detected, because of data quality issues. It's one of the most critical factors. I mean, monitoring fields like transaction data is vital, but it's incredibly challenging. For instance, when you have a system processing billions of transactions every hour, it's really hard to check each one for missing information. I would suggest doing something similar to what we discussed for the first principle, monitoring fraud rates. If, for example, they suddenly stop sending the device ID, and the fraud detection system stops catching certain cases, you'd notice that. So, by monitoring these rates as we talked about earlier, we could likely address this issue as well. But yes, this principle is hugely important. From a data science perspective, the biggest enemy is often data quality. #00:06:04#

Interviewer The next design principle is the hybrid detection approach. This principle focuses on enhancing detection efficiency and adaptability for compliance officers in the context of operational needs. It advocates for employing a hybrid approach that integrates both AI and rule-based methods. This combination provides flexibility in adjusting detection rules and precision in fraud detection, leveraging the strengths of both methodologies to improve accuracy and reduce the manual review workload. The idea is that this approach ensures the system remains effective and responsive to evolving fraud patterns while allowing for timely adjustments to changing requirements. What I'm suggesting is that it's probably not ideal to rely solely on an AI-based or rule-based approach. Instead, a balanced combination of both is the way to go. Some companies might prefer a more rule-based approach with AI to reduce manual work, while others might lean more towards AI. But ultimately, a balanced approach that incorporates both methods is the correct way to go. What do you think about this design principle? #00:07:20#

Interviewee I think hybrid systems are most effective when used wisely. If you have a hybrid system where a model triggers some cases and then rules start spamming, the overall efficiency drops. What I like about rules is that they should often be applied only to edge cases, situations where we already know they won't generate many cases. You should expect a rule to trigger just a handful of cases per day, focusing on something really uncommon. It's important to detect early when a rule is becoming noisy and when that risk factor needs to be incorporated into the model. Typically, the AI model runs on certain factors, but it might not include something like crypto. For instance, if a customer onboards using crypto, maybe we don't include that in the model initially, which is why we create a rule. But when the rule becomes noisy, that's when we should leverage it for the model. Otherwise, we start missing things, and the efficiency of the system drops. #00:08:34#

Interviewer The next design principle is scalability and flexibility. To maintain long-term usability and effectiveness for compliance officers, especially in the context of increasing transaction volumes and evolving fraud techniques, it's crucial to employ scalable infrastructure, modular system design, and regular performance assessments. This approach ensures that the fraud detection system can efficiently handle growth, adapt to new fraud methodologies, and remain effective over time. What do you think about the importance of scalability and flexibility in such a system? #00:09:53#

Interviewee It's the dream of every data scientist and engineer, right? Having a system that scales up without headaches. It's definitely achievable, but it requires a lot of work. Scalability is absolutely a requirement, there's really nothing more to add. The system needs to be able to handle loads, like on Black Friday, when transaction volumes can double or even triple. If the system isn't scalable, it could break under the pressure, and then you miss a lot of transactions. So, yeah, it's definitely a requirement. #00:10:35#

Interviewer And then the next one is design principle six, which focuses on dynamic and interactive explainability. We've discussed this a lot. The goal here is to enable dynamic exploration of data and provide interactive explainability features for compliance officers using the fraud detection system. This means offering user-friendly, interpretable insights, clear reasoning paths, and confidence levels in the system's decisions. This approach enhances trust, allows users to understand and validate the system's outputs, and supports effective decision-making in fraud cases. What do you think about this aspect or design principle? #00:11:11#

Interviewee This principle is about giving the user the information they need, right? For instance, if they need to view some plots or review the history of a customer, this feature is definitely helpful. I wouldn't say it's mandatory, but it helps a lot. Most importantly, and something I often see lacking in many systems, is that the user should be able to see all the information the system has. If there are new data points or fields for a transaction, the user needs to be able to access them directly within the fraud detection system. If they have to jump to another platform, it diverges the tools and makes the process less efficient. So, while interactive explainability might not be mandatory, it's really supportive for the decision-making process of the fraud expert. Without it, they're not seeing the whole picture. #00:11:59#

Interviewer So, what I meant before is that the user shouldn't have to leave the platform or the system. For example, they shouldn't need to go to Google to search for something or access another platform to get the

information they need. Everything should be accessible within the fraud detection system itself. #00:13:20#

Interviewee Let's say you're a payment processor, right? You process transactions, and some are sent to the fraud detection system. However, maybe the fraud detection system doesn't show all the data points. For example, it might not display the conversion rate of the currency used in a transaction, even though it uses that information internally. When the fraud expert is investigating the transaction, they might need to know the conversion rate for that day. If the system doesn't provide this information, the expert would have to switch to another platform to find it. This is where problems can arise, the data might diverge. One platform might show one value, and the other might show something different, or the conversion rate might only be available for the current day, making it impossible to investigate cases from a month ago. #00:13:30#

Interviewer That's interesting, of course. For example, if someone pays a large amount in a foreign currency, like 10'000 but in Swiss francs, it ends up being just 500 bucks. #00:14:31#

Interviewee Exactly. #00:14:39#

Interviewer You might think that 500 bucks isn't a big amount, and decide to leave it, but in the other country, it's a lot of money, right? That's where it gets really interesting. #00:14:40#

Interviewee For instance, with the Japanese currency, if you look at the past two years, it has drastically dropped compared to the euro. If you compare the values from now and before, they tell completely different stories. For example, €100 before was something significant in Japan, but now it's a completely different situation. So, yeah, this data must always be available to ensure accurate analysis. #00:14:47#

Interviewer That's true. Thank you so much. Now, the next one, the seventh design principle, is about integrated preventive and detective measures. What I mean here is enabling a strategic shift to combine both preventive and detective fraud management, not just for managers but also for compliance officers, within the context of comprehensive fraud protection. It's important to employ fraud detection algorithms that support both preventive and detective measures. This approach allows the system to use the same metric model, ensuring more secure protection for both card users and the company. For us at amnis, we're starting with a more detective approach because we don't have that much data at the moment. But strategically, to protect our customers and the company, it's important that as our models become better trained and we gather more data, we shift from only a detective method towards a more preventive approach as well. What do you think about this principle? Do you agree that it's important not just to have both preventive and detective measures but to integrate both approaches in the long term? #00:15:20#

Interviewee For preventive measures, something really important is to ensure that the integration is up to date. Amnis has been operating for a few years now, and during that time, new functionalities have likely been added to fraud detection solutions that may not have been integrated with amnis, simply because they were developed afterward or only partially revisited. This means that some new data points, like addresses or IPs, which weren't considered fraud or risk factors before, might now be relevant due to emerging trends. So, I would emphasize the importance of keeping the system up to date. This doesn't just mean checking newsletters, but really staying informed about which risk factors are now being handled by the fraud protection system. #00:16:41#

Interviewer Is there anything else you need to consider when switching from a detective to a preventive approach? Or do you think it's as simple as just using the same algorithms and saying, OK, now just apply them to incoming transactions instead of approved ones? I mean, is the switch really that easy in the end, or is there much more to consider? #00:17:56#

Interviewee With preventive measures, you always have the challenge of detecting the transaction before it happens or before the funds are moved. That's always more labor-intensive. With detective measures, you can perform it retroactively, so you usually have more data to work with. For example, with detective methods, you can analyze the previous week's transactions to identify fraud because you have the entire history to look at, it's like having a view of the past to understand the future. We actually do both at the same time because, for every transaction, we compare it with the entire profile of the accounts or customers. So, we don't really have that switch you mentioned because we're already handling it on a transaction level. #00:18:23#

Interviewer The last one is about bias mitigation and ethical decision-making. This principle focuses on ensuring fair treatment of all card users and minimizing human bias in the decision-making process, especially for compliance officers. It's important to employ algorithms that detect and measure bias, perform continuous monitoring, and conduct regular audits and updates. This approach ensures that decisions are made without undue bias, fostering trust and reliability in the fraud detection system. Additionally, it's not just about bias within the system itself but also about bias that might be triggered by a compliance officer. For example, if a compliance officer labels an obviously fraudulent case as not

being fraud or doesn't work correctly, there's a risk of introducing bias through human error as well. What do you think about this design principle? #00:19:29#

Interviewee I'll give you two points of view. First, from a data scientist's perspective, the objective should be to find the best possible model, right? Independently of what we use. However, that can lead to using data that might become unethical. For example, using names that could imply ethnicity wouldn't be fair. But I think these constraints should already be established by product design before the data scientist even starts tuning the model. Let's say, before a data scientist puts in a lot of effort preparing features and so on, there should already be a clear list of what's off-limits. For example, maybe we're told not to use names, which we definitely don't use, but also maybe not to use city names either. Some cities might have lower income levels where fraud might occur more frequently, and if the model uses that information, it could end up discriminating against those cities, even if it improves the fraud detection rate. So, while the model might be more effective, it could be unethical. But I believe this shouldn't be the data scientist's decision because there could be a conflict of interest. On one hand, I want the best model, but on the other hand, I might consider using names unethical while thinking that using city names is fine. Therefore, these constraints should be defined by someone else, like the risk officer of the institution, to allow data scientists to focus on the actual objective of their task without ethical conflicts. #00:20:28#

Interviewer Yes, that makes complete sense. #00:22:19#

Interviewee And I'd say in that case, the user of the platform should always have the last word. AI systems or machine learning systems might always be slightly biased, just like humans are. For example, they might be biased based on data, just as humans can be based on personal experiences. So, it's the responsibility of the institution using these systems to ensure that humans always have the final say. We shouldn't blindly trust the system just because it's AI. We should always apply human judgment, being rational and saying, "OK, that's fraud because of specific reasons, and not simply because the person's country of origin is X. Human oversight and rational decision-making should always be present. #00:22:29#

Interviewer Thank you so much. Now, in general, do you think these principles cover all the critical aspects of an AI-based fraud detection system? And do you have any final suggestions for improvement? What do you think might be missing or should be added? #00:23:27#

Interviewee I think these principles provide a pretty solid guide for developing an AI-based fraud detection system, to be honest. As I'm browsing through them, I don't see anything major that's missing. We've talked about continuous improvement, the user-centric piece, feedback loops, and data quality. I don't think we're leaving anything out. I often mention the idea of controlling the beast, and in this case, the beast is the AI model. We need to monitor it, ensure we're not introducing bias, and keep everything under control. One possible addition could be on the prevention side, we could include training for both data scientists and product teams. Not to teach them how to perform fraud, but to help them stay updated on the latest trends in fraud. That knowledge could be really helpful in developing these systems. Maybe we could include this in design principle seven. #00:23:43#

Interviewer Exactly. If you don't fully understand what fraud is and how it can happen, it's hard to work optimally with fraud cases. You can't just tell someone to look at a transaction if they have no clue what fraud is or what they're looking for. They might see a transaction and think, what is this? while the other three are correct. So, training is definitely a good point to mention as well. #00:25:11#

Interviewee Right. For instance, a data scientist might never think of using SMS features if they don't know that phishing can occur via SMS. #00:25:41#

Interviewer Exactly. Or vishing. #00:25:54#

Interviewee If you don't know that SMS is vulnerable, then you might not even consider how someone could get scammed via SMS, for instance. #00:26:03#

Interviewer That's good. OK, perfect. Thank you so much. I think that's everything. Do you have any other suggestions, or should we stop here? #00:26:12#

Interviewee No, I'm good. I think we went through everything thoroughly. #00:26:22#

Expert 3 at Hawk

Date: Auguste 9th, 2024

Time: 15:15

Place: Microsoft Teams

00:00:00 – 00:24:22

Interviewer Here, it's really about getting your feedback and your thoughts on these principles. I want to know what you think about them. As I said before, feel free to be really critical. Let's start with the first one. The first design principle is about adaptive learning and continuous improvements. To improve fraud detection accuracy and performance over time for end users like us, especially in the context of evolving transactional behaviors and changing card user habits, it's important to employ real-time learning mechanisms, analyze historical data, and continuously integrate feedback from compliance officers. For example, this involves implementing constant updates and refinement processes that dynamically adapt to both emerging fraud trends and normal variations in card user behaviors. This approach ensures that the fraud detection system remains accurately tuned to fraudulent activities and legitimate user trends, enhancing the system's reliability in a rapidly changing environment. What do you think about this first design principle? #00:00:13#

Interviewee I think this is very crucial. Actually, this is a requirement by nearly all of our customers. They typically require retraining every six months or one year to update the model and deploy it again. The main reason for retraining is, of course, to detect new fraud patterns, since the old patterns are already covered by the existing model. So, not only is it crucial, I would say it's a requirement by customers at this point. #00:01:25#

Interviewer By the way, I know it's quite complicatedly written, but it's based on a scheme proposed by literature and researchers for how design principles should be formulated. So they might sound really theoretical. #00:01:51#

Interviewee Yes, I understand. #00:02:08#

Interviewer Thanks. So the next one is user-centric development and alignment. This principle focuses on creating a tailored fraud detection system that aligns with the business model, goals, services, and customer base for managers and compliance officers. During the development process, it's important to actively involve end users and establish regular feedback loops to tailor the system to specific business needs. This approach ensures that the fraud detection system is effectively aligned with strategic objectives and optimizes the cost-benefit ratio for each end user by meeting specific goals, such as reducing manual work, increasing efficiency or speed, enhancing risk management, and making optimal use of available resources. For example, at amnis, like us, we don't have that many human resources, so generating a lot of alerts might not make much sense. Whereas a company like Citigroup, which has a large department solely focused on fraud, can handle more tickets being created. So, there isn't one perfect fraud detection system, it really needs to be tailored for each end user. What do you think about this design principle? #00:02:17#

Interviewee I mean, if that wasn't the case, we would just hire a couple of data scientists, build one model using the same features for every client, and deploy it. That would make our lives very easy, but that's not the reality. For example, we had one client who dealt primarily with trucks. I'm not sure if he told you about that, but OK. In that case, we had to focus entirely on latitude and longitude, like if a truck driver covered a large distance in a short amount of time or something like that. That's not something we normally use. Similarly, we have a new customer from South Africa with specific account types that we don't have here. They only deal with mobile phones, so we had to take that into account and monitor nighttime transactions, which can be related to fraud. So, yeah, that's entirely true. #00:03:41#

Interviewer The next one is something we've discussed extensively before, especially in our last conversation, which is comprehensive data integration and quality assurance. This principle is about enhancing fraud detection accuracy by focusing on data quality and integrity. It involves thorough data understanding, comprehensive preprocessing techniques, advanced feature engineering, secure data handling, and the integration and analysis of various contextual data points. It's not just about the transaction data itself, but also about factors like time, location, logging behaviors, device IDs, and IP addresses. This approach ensures that the model is built on a robust and reliable data foundation, accurately identifying fraudulent activities. What do you think about this principle? #00:04:55#

Interviewee Some of these factors are very critical, like time. Time is definitely crucial. As for the others, I wouldn't say they're all absolutely essential. For example, we've been dealing with IP addresses for a while now, and we're just starting to integrate them. But we had a U.S. customer where we did a POC, and in that case, the IP address was very helpful. It allowed us to see if 10 different accounts were logging in from the same IP address and withdrawing money from the same ATM, which indicated something suspicious was going on. So, while IP addresses are obviously helpful, it doesn't mean we couldn't function without them. But there are some fields, like time, counterparty, or country to some extent, that we really can't do without. Not all of these factors are essential, but they're definitely helpful. #00:05:55#

Interviewer These are just examples, but obviously, this list is endless, especially as more and more features become available. There are increasingly more ways to gather data beyond just the transactional information, so this list could become endless in the future. You'll have to pick the most fitting or helpful ones, and that selection process will likely be guided by you. Now, moving on to the next one, which is the hybrid detection approach. #00:07:00#

Interviewee Exactly. #00:07:26#

Interviewer This principle is about enhancing detection efficiency and adaptability for compliance users by employing a hybrid approach that integrates AI and rule-based methods. This combination provides flexibility in adjusting detection rules and precision in fraud detection, leveraging the strengths of both methodologies to improve accuracy and reduce manual workloads. This approach ensures that the system remains effective and responsive to evolving fraud patterns and allows for timely adjustments to changing requirements. For example, if a new country is blacklisted or becomes high risk, you can quickly implement that as a rule, while AI excels at learning and adapting to customer behaviour. By leveraging the best of both approaches, you ensure the system is both flexible and robust. What do you think about this? #00:07:32#

Interviewee I mean, definitely. More specifically, if you're, let's say, training an unsupervised model, which we usually do for anomaly detection, there are situations where we don't have labels. In those cases, we can't train the model to assign a 90% anomaly score whenever a transaction is made to or from Russia. Without labels, we can't train the model to do that effectively. In those cases, rules are definitely very important, especially for binary situations like flagging every crypto transaction or something similar. You can't always rely on even a supervised model to consistently catch that, so rules play a crucial role. #00:08:30#

Interviewer But do you think in the future we'll rely solely on AI-based models, or will there always be some rules implemented? #00:09:15#

Interviewee It really depends on compliance requirements, which can be very strict. For example, there's a requirement by European banks, I think you might know more about this, that every transaction greater than €10'000 should be flagged. We can't rely solely on a model to ensure that, although there are some mathematical ways to do it, they don't work with every model. So, rules are necessary in these cases. #00:09:22#

Interviewer I mean, in the end, it makes things more complicated if we try to avoid rules altogether. Implementing some rule-based sets actually makes life easier for you guys and for the system overall. #00:09:50#

Interviewee I mean, I think in the future, rules will always be there, at least for the critical ones. AI will continue to play a big role, but rules will still be necessary. #00:09:59#

Interviewer The next one is scalability and flexibility. This principle is about maintaining long-term usability and effectiveness for compliance officers, particularly in the context of increasing transaction volumes and, consequently, the increasing amount of data and evolving fraud techniques. It emphasizes the need to employ scalable infrastructure, modular system design, and regular performance assessments. This ensures that the system can efficiently handle growth, adapt to new fraud methodologies, and remain effective over time. What do you think about this design principle? #00:10:15#

Interviewee I very, very much agree. From what I understand, you're talking about how easily the system can handle a situation where the client starts sending twice the amount of transactions because they've grown, and whether we can manage that increase. I also think this ties back to your first point, how easily we can update the model if fraud patterns change. It's definitely crucial. We can't just assume that if a client is currently sending 100 transactions per second, it will stay that way. Clients can grow quite rapidly, so scalability is very important. #00:10:49#

Interviewer So, would you recommend combining it with the first principle since they're related, or do you think it should be kept separate at this point? #00:11:27#

Interviewee I mean, the first one was mostly related to fraud patterns, while this one is more about development and operations, DevOps. I'm not sure if you can combine both because this one is also talking about the volume of transactions, the sheer mass you receive, and whether you'll be able to handle that. #00:11:38#

Interviewer Well, I mean, generally speaking, if we're now growing and you have twice the customer base, you're not only sending 100% more transactions, I don't know the exact numbers, but it could be duplicated, and then the system might not handle it anymore. I guess this becomes even more critical when you have a preventive approach where you need to block or approve transactions in real time. In that case, scalability and flexibility become crucial parts of the system's success, correct? #00:11:52#

Interviewee Exactly, exactly. In our case, we just add more nodes as the customer's transaction volume increases, so I would say we're very flexible. #00:12:30#

Interviewer The next principle is about the interaction with the system. The goal here is to enable dynamic exploration of data and provide interactive explainability features for compliance officers using the system. This means offering user-friendly, interpretable insights, clear reasoning paths, and confidence levels in the system's decisions. This approach enhances trust, allows users to understand and validate the system's outputs, and supports effective decision-making when handling fraud cases. For example, we had an approach where users could change a feature, like the amount or location, and see how the system's output would change. This allows for interactive exploration, helping users understand how the system behaves and how the output is generated. It's not just playful, it's more about giving users the ability to see and understand how the system works and what happens when certain variables change. What do you think about this design principle? #00:12:41#

Interviewee Interactive explainability, in my opinion, might be overkill. It's very useful for research purposes. For example, during my master's, we often used interactive explainability to see what would happen if we changed a feature, like which part of an image was crucial in determining that it was a car and not a person, or what happens if you modify certain features in sound or images. In this field, however, I doubt that an investigator on the other side would be doing that or would even be interested in it. Compliance might use it once a year, just to ensure that nothing is going wrong, but otherwise, I think general explanations are very important for building trust. But going overboard on it might not be necessary. For instance, right now, we're considering extending our explanations a bit more, currently, they're on a transaction level, but we're thinking about generalizing them to subcategories, like gender or other attributes. This is interesting for us as developers because we understand the system, but I doubt an investigator or someone in compliance would be interested in that level of detail. It's more likely to appeal to someone who is curious, but not for compliance purposes, I guess. #00:14:09#

Interviewer That's a really good point. It could be useful for educational purposes, like when you have new employees who need to get up to speed. You could use it to educate them on how the system works and how the outputs are generated. It could be a valuable training tool in that context, but it might not be as practical for daily work. #00:15:39#

Interviewee Exactly. #00:16:00#

Interviewer So, they might play around and see, what happens now? Is it going to be flagged as fraud or not? That's completely understandable. Good point, thank you. The next design principle is about integrated preventive and detective measures. What I'm trying to convey here is the importance of combining both preventive and detective approaches. Preventive measures involve blocking or accepting transactions before they're even approved, while detective measures involve checking all transactions that have already been approved. In the context of comprehensive fraud protection, it's important to employ fraud detection algorithms that support both preventive and detective measures. This allows the system to use the same well-trained model, ensuring more secure protection for both card users and the company. In the long term, it doesn't make sense to rely solely on either preventive or detective measures. The best approach is to integrate both, ensuring that you can protect your customers and the company more effectively. What do you think about this design principle? #00:16:03#

Interviewee So, by preventive, you mean stopping the transaction, and the other one is...? #00:17:25#

Interviewer Exactly. The other one is for all approved transactions, after the money is already gone, you check them again to see if there was a fraud case. #00:17:34#

Interviewee It's just alerted but didn't block the transaction, right? I mean, that's the idea, that these two processes happen together, and I think they should. For example, if money is going to a sanctioned country like Russia, I'm just using Russia as an example because, since I joined, it's been a constant focus due to the war, there should be a rule that blocks the transaction immediately. In other cases, like when focusing on round amounts, you wouldn't block the transaction automatically. Instead, you should have an alert that flags it for later investigation. For instance, if someone sends \$1'000 transactions to 10 different people

within 10 minutes, that should be investigated afterward. So, it's important that these processes work together like this. #00:17:45#

Interviewer I always say that for us at amnis, we have preventive measures in place, like the rules you mentioned for sanctioned countries. But the full model is currently working more in a detective mode, where we go through all the accepted transactions. I may need to formulate this a bit better because the emphasis should be on the idea that, in the long term, we should also have a working model that, once it's well-trained and there's enough data, can be used to block transactions before they're even approved. The idea is to check beforehand if something is fraud, and I may need to express this a little better. And the last one is bias mitigation and ethical decision-making. To ensure fair treatment of all card users and minimize human bias in the decision-making process, it's important to employ algorithms that detect and measure bias, perform continuous monitoring, and conduct regular audits and updates. This approach ensures that decisions are made without undue bias, fostering trust and reliability in the fraud detection system. This principle also covers bias that might be introduced by compliance officers. For example, if an officer is lazy or doesn't thoroughly check alerts and just dismisses cases as not being fraud when they actually are, the model could learn from these incorrectly labelled cases and become biased as a result. So, this principle addresses both bias in training and bias introduced through user interaction. What do you think about this design principle? #00:18:44#

Interviewee I mean, I think this is very important, right? But I've had so many sessions about this, how do we tackle it? For example, we have to use country as a feature because there are blocked countries and high-risk areas, so we check if transactions are made to or from these places. But let's say a person was born in Russia, and now the model is already biased against them because a lot of illegal activities happen from high-risk countries. So, we can't easily avoid using this feature. However, if we were given a nationality feature or race, like whether the person is black, brown, or white, we ethically cannot use that feature. What would the explanation be if a transaction was blocked because the person was black? That would ruin the company's reputation. So, it's a bit hypocritical that we can use country but not race. We try to lower bias at the race level, but it still exists at the country level. There will always be some bias, we just try to reduce it as much as possible. In the end, it's about generalizing how transactions happen, so some bias will always be there. We try to minimize it using explanations. If we weren't using explanations, we might have included nationality as a feature, but that's unlikely because no one would see that the transaction was blocked due to the person's race. #00:20:32#

Interviewer Where would you even get the information that someone is of a certain race? I mean, it's probably not in the transaction data. #00:21:50#

Interviewee Exactly. I mean, it's the same with gender or KYC data. #00:21:58#

Interviewer That's going to be a challenge. But regarding the ethical part, maybe we can move on here. I guess we discussed it earlier, how bias can also affect the customer. For example, if you have transactions made by large companies or crypto transactions, the system might be biased because your customer base is mostly small to medium-sized enterprises. But then, when you get crypto transactions, it doesn't work as expected. I also wanted to discuss that here under bias mitigation. #00:22:06#

Interviewee You're right. In that case, the best practice would be to train a different model specifically for crypto transactions so it's not biased at all, it's just focused on crypto. The same goes for gender. If you know the population is roughly 50/50, you could train different models accordingly. We already do that for individual customers versus organizational customers because their transaction patterns are so different. So, we can also apply this approach to different genders or specific types of transactions like crypto. #00:22:50#

Interviewer Maybe, as I said, the ethical part can be ignored here. It's more related to bias and could be formulated differently. So, with that in mind, here's my last question: Do you think these principles cover all the critical aspects of an AI-based fraud detection system? How do you feel about them overall? Do you have any general feedback? Is there anything missing? #00:23:20#

Interviewee I think they were quite good and they do cover the critical aspects. If anything, there are more technical considerations, like when you want to change the model, but that's already covered by the relearning and continuous improvement processes. But again, that's more of a technical issue. Overall, I think that's it. #00:23:45#

User 1 at amnis

Date: July 23rd, 2024

Time: 16:30

Place: Microsoft Teams

00:00:00 – 00:44:09

Interviewer Now, this interview is about evaluating my derived design principles. I really want to get your feedback on them, how you like them, what you think about them, and especially how efficient or effective you think these design principles could be in providing a framework for AI-based fraud detection systems. I'd like to start with the first design principle that I derived from the literature and user feedback, which is the adaptive learning and continuous improvement principle. This principle is for developers to improve fraud detection accuracy and performance over time for compliance officers like you. In the context of evolving transactional behaviors and changing card user habits, it's important to employ real-time learning mechanisms, analyses historical data, and continuously integrate user feedback, like from you when you say, that was fraud or that wasn't fraud. It also involves implementing constant updates and refinement processes. So, when a fraud detection system is deployed, it's not the end of the work for developers, it still needs to be monitored and updated. These mechanisms help the system dynamically adapt, not only to emerging fraud patterns but also to normal variations in card user behaviors. This approach ensures that the fraud detection system remains effectively tuned to both fraudulent activities and legitimate user trends, thereby enhancing the system's accuracy and reliability in a rapidly changing environment. So, that's the first principle. I know it's complicatedly formulated, but I just want to know how well you think this design principle will ensure that the fraud detection system remains effective over time. Do you have any suggestions for improving this principle? What are your thoughts on it? #00:00:05#

Interviewee Actually, this is it, right? That's the reality. As you said, the monitoring and the detection will always be a step behind because criminals are constantly ahead. So, this principle of continuous improvement and adapting to new patterns is really the only way to effectively spot misuse or criminal activity. Technically, there can't be anything different, this is the reality. I can't see anything that needs to be changed here. It makes total sense. #00:02:15#

Interviewer Maybe how it's written, do you think it's complicated, or is it clear what's meant? Did you understand what I was trying to convey with this design principle? #00:03:17#

Interviewee Actually, of course, you'd need to add commentary to explain what it means, especially if you're talking to non-subject matter experts, which will definitely be the case at times. If the person isn't a senior in such an environment, it could be perceived as overcomplicated. But I don't feel there's anything inconsistent or that requires simplification. It's really difficult to say that anything is missing, the setup is clear. I'd keep it as is, but with the understanding that you'll provide explanations as needed. Fraud is never easy and is really dynamic. As you correctly mentioned, we're in a rapidly changing environment, with new patterns being developed by fraudsters almost every week. That's just the reality we're dealing with in modern fraud detection. #00:03:25#

Interviewer Let's move to the next design principle. So, we talked about learning, now it's also important to consider customer involvement and business strategy adaptation. What we want to convey here is that developers should create a tailored fraud detection system that aligns with the business model, goals, and services of each client. For example, we have completely different clients than a private bank might have, and the customer base for managers and compliance officers like you is different. In the context of the fraud detection system development process, it's important that there's active involvement with end users like you or the management, and that collaboration is maintained throughout the development process. Regular feedback loops are also essential. Similar to what's happening now with our fraud system, there's always communication between compliance, IT, and the developers. This approach ensures that the fraud detection system is effectively aligned with strategic objectives, optimizing the cost-benefit ratio for each end user by meeting specific goals, such as reducing manual work, increasing efficiency or speed, or whatever our goals are. It also makes optimal use of the available resources that your compliance department or the overall company has. What do you think about this principle? Does it sufficiently address the need for alignment with business goals and user needs, or how could it be enhanced? What are your thoughts on this design principle? #00:05:05#

Interviewee If you go to the core of it, the fundamental principle of this whole anti-fraud exercise is a satisfied customer. The customer is central to everything because we're trying to protect them against criminals, fraud, and other risks. That's the basic principle, and it's what we need to support. We have limited

resources, just two hands, and that's why we need the most efficient system possible to cover fraud detection. Technically, it's way more complicated, but the basic principle is customer awareness and involvement. This remains the main pillar of the whole exercise. If this fails, if the customer fails in their role, then the first level of anti-fraud protection becomes much more complicated. The fraud detection system will still spot certain behaviors, that's clear, but the customer plays a crucial role in both safety and detection. Mutual collaboration between the system and the customer makes the whole process more effective. That's the overall idea. #00:06:49#

Interviewer What you mean is that the main objective, to protect the customer, our customer, is missing here, and the focus is too much on the management side? #00:08:45#

Interviewee Exactly, because that's what really matters, right? On our side, whether it's through AI, manual processes, or whatever we do, we are ultimately providing a service, payments and cards, for the customer to use. If the customer isn't disciplined or safe, then misuse is inevitable, and criminals will take advantage of such vulnerabilities. So, yeah, it's crucial not to forget about this core principle when discussing fraud detection and anti-fraud strategies. The customer plays the key role, and this is what it's all about. #00:09:16#

Interviewer Perfect. That makes complete sense. The aspect of focusing on our customer is missing, and I've noted it down. Thank you so much. Let's move on to the next design principle, which is more technical. This one is about comprehensive data integration and quality assurance. For developers, the goal here is to enhance fraud detection accuracy for you, the compliance team, in the context of data quality and, especially, data integrity. It's important to employ thorough data understanding, knowing what we have, comprehensive preprocessing techniques, advanced feature engineering, secure data handling, and integration analysis of various contextual data points. It's crucial that we don't just focus on the transaction itself but also consider factors like time, location, login behaviours, if we have that data, device IDs, and IP addresses. This broader perspective ensures that the model is built on a robust and reliable data foundation, which helps in accurately identifying fraudulent activities. What do you think about this principle? #00:10:30#

Interviewee Yes, again, to safeguard our customers, we use everything we have, our infrastructure, experience, quality assurance processes, and lessons learned from previous fraud cases. We try to build the system to be as preventive and vigilant as possible, to spot anything suspicious and trigger immediate alerts. The goal is always to safeguard the customer and keep them happy because that's the perceived benefit and added value we offer. That's why companies have a 24/7 anti-fraud helpline. As a fintech company, we want to use the machine as much as possible to minimize manual intervention. However, sometimes it can be overkill or counterproductive. For example, if someone suddenly travels to Thailand and wants to pay for a hotel, the system might block the transaction because it thinks the person should be in Switzerland, where they usually stay. The system's preventive measures might see this as suspicious. But that's something we can't always avoid. The key is customer satisfaction. If it's a weekend and the customer can't resolve the situation because the system blocked their payment, that's a problem. We can try to minimize such exposure by gathering more details, like upcoming trips, but it's never perfect. So, there's really no better way than to follow this approach with quality assurance and vigilant detection. That's the reality, and we have to work within it. #00:11:45#

Interviewer Thank you so much. Now, onto the next design principle, which is all about the approach itself. This one is called the balanced detection approach. To improve detection efficiency while also minimizing false positives for compliance officers, especially in the context of operational efficiency and cost-effectiveness, it's important to integrate both AI and rule-based methods. This combination provides flexibility and precision in fraud detection and reduces the manual review workload. What I'm trying to say here is that while AI provides a good solution for minimizing false positives, like auto-closing or auto-detecting certain behaviors or card user patterns, rule-based methods are also necessary. Rules allow us to incorporate specific criteria, like excluding high-risk countries or setting amount limits. Using only static rules isn't the right approach, but relying solely on AI isn't either. It's about finding the right combination of both, rule-based and AI. What do you think about this design principle? #00:15:27#

Interviewee Totally, totally agree. Nowadays, the real scenario is that we don't use AI at all for card monitoring, and we can demonstrate every day that the volume of card alerts, whether they're fraud prevention alerts or transaction monitoring, AML, category alerts, brings us to situations where it's clear that certain alerts are unnecessary. For example, yesterday, we closed an alert for a transaction in Las Vegas where a person was paying for a hotel. It was so obvious that the person was actually there, based on their activity, and there was no indication of fraud. But the system still generated an alert just because the person is from Switzerland and was using their card in Las Vegas. The frequency of their activities increased, and this triggered more alerts, even though everything was fine. This is where AI could be really beneficial, helping us get rid of these unnecessary alerts. For instance, if an alert shows that the card was used in Las Vegas and then, on the same day, in Zurich at the Hauptbahnhof, that would indicate something might be wrong because the person couldn't have travelled back to Zurich so quickly. This is the kind of scenario where AI could make a big difference. We're still trying to activate AI for cards, but it's subject to discussion. The goal is to focus on relevant alerts. Yes, from a fraud perspective, we don't want to allow

even 10 Swiss francs to be stolen, but there's a difference in risk between that and, say, 100'000 francs being stolen. We handle different risk categories and attributes. The combination of monitoring rules for fraud and money laundering prevention is something we already have in place. Whether it's good or bad, we don't differentiate much between preventing money laundering or stopping criminals from stealing money. It's all about spotting and preventing criminal activity. So, while manual, fixed-rule setups are the very basic and still functional, it's clear that AI is something we want to benefit from. It can save us effort, capacity, and help us maintain our brand. #00:16:52#

Interviewer Thank you. Now, onto the next design principle, which is about scalability and flexibility. This principle emphasizes the importance of maintaining long-term usability and effectiveness for compliance officers, especially in the context of increasing transaction volumes and evolving fraud techniques. It's crucial to design the system for scalability and ensure its flexibility. For example, if we onboard new clients or introduce new services, the system should support modular updates and expansions. This approach ensures that the fraud detection system can handle growth and adapt to new fraud methodologies, maintaining its effectiveness and usability over time. What do you think about this design principle? #00:22:50#

Interviewee Yeah, that's exactly it. This is a summary of everything we've just covered, right? Every compliance officer or surveillance and monitoring process aims to see relevant and appropriate alerts. In an ideal world, we want to see clear fraud alerts or money laundering alerts. However, the reality is quite different. We find that 99.9% of the alerts are false positives, and that's the challenge. We're constantly trying to eliminate the noise, and that can only be done through a proper setup, regularly enhanced and modified, while clearly monitoring the methodologies. That's the only way to manage it effectively. When it comes to new customers, we often know nothing about them initially. They might declare their intentions, like a customer from the Netherlands saying they'll only use their card for travel within Holland. But the next day, they're on a plane to Dubai. It's not a problem, but the KYC factor and fraud prevention measures kick in. Sometimes, it leads to misunderstandings, like the system flagging the trip to Dubai as suspicious because the customer didn't mention it. This highlights the complexity of managing such situations. The only measures we can apply are to set up the most efficient and possible rules to create patterns for a given customer base, or specifically for each customer. It's really a case-by-case scenario, and you never know what's going to happen next. That's why we have alerts, because customers behave unpredictably or, at least, in ways we perceive as unpredictable. This also leads to the never-ending tension between business sales and the compliance side, like AML monitoring and fraud detection. The sales team might not always understand why we need certain information, or they may not communicate it effectively. So, these principles you've described are really spot on. They encapsulate everything we can say about managing these challenges. #00:23:45#

Interviewer Do you think this design principle is more or less the same as the others, just summarizing what's already clear in the previous design principles? So, maybe we don't really need to separate scalability and flexibility as a standalone principle because it's already embedded in all the other principles we've discussed. Do you think this could be removed, given that scalability and flexibility are inherent in the other principles? #00:27:58#

Interviewee It's clear. Scalability is where we want to be. I'm always trying to explain that scalability is really nice, but it's not always possible to implement. So, I do understand why this chapter or subject is specifically here. Let's keep it, it's not duplicating anything. It's specifically targeting the most efficient way to achieve scalability and how to use flexible methods to make that happen. The only way we can do that is by modifying the parameters appropriately so we can stay vigilant and maintain control, ensuring we have an efficient monitoring platform. And that's exactly how it should work. #00:28:27#

Interviewer Now, onto the next one, which might be especially interesting for you, it's about dynamic and interactive explainability of such systems. This principle focuses on helping users understand and validate the system's output and supports effective decision-making in handling fraud cases. What do you think about this design principle? Do you have any input on it? #00:30:04#

Interviewee This principle is clearly always required by regulators, auditors, and those reviewing the effectiveness of the monitoring system. It's a crucial principle because people in audit functions need to easily understand what we're trying to achieve with our monitoring rules, how the system works, what it's covering, and which principles are being addressed. It has to be crisp and simple. That's exactly how the monitoring setup should be established and explained. My understanding of AI's evolution is that much of the hidden magic is left with the machine. For example, I learned from discussions with an Israeli AI expertise company, which is highly valued in the field. They suggested, give us all the transaction activities, and our system will spot the issues. But that's where the problem lies, letting the machine decide everything. While the machine might identify specific problematic patterns in transactional activity, the real challenge is understanding and justifying the machine's decisions. Regulators and auditors dislike situations where the process is opaque, where we say, this is the machine, we trust it, and we just look at the alerts it creates. The machine might provide a justification for why an alert was generated, but then it's up to a human reviewer to decide if the machine was right or if it's a false positive. This interaction between the human and the machine is crucial. However, relying solely on the machine is risky and problematic. From

my experience, it's not a good way to support your monitoring system in front of auditors, you're likely to fail if you can't explain what the machine is doing. You'll end up with many findings due to a lack of appropriate understanding of the setup. But if you're strong enough to understand the logic behind AI and all the parameters involved, and you can explain this functionality to a human auditor or regulator, then you're in a winning position. Personally, as we move towards more AI-driven activities at Hawk, especially with cards, I know we still need to manage the technical setup in a way that aligns with traditional audit principles. The balance is crucial. While we're embracing AI, we must ensure we understand how the system and AI operate. That's the key, the core of what we need to focus on. #00:30:50#

Interviewer The seventh principle is proactive fraud prevention. With this, I want to emphasize the need for managers to strategically shift from a reactive, more detective approach to proactive fraud management. This shift is crucial for providing more secure protection for both card users and the company itself, safeguarding against reputational or financial damage. It's important to develop advanced predictive analytics capabilities within the fraud detection system so it can identify and prevent potential fraud before it occurs. What I'm getting at is that the system must be capable of switching from a detective approach to a preventive approach. What do you think about this design principle? #00:37:34#

Interviewee Definitely yes. And just an idea, with this proactive fraud prevention, which in my view represents 50% of the fraud monitoring system, I would actually start with this setup. I would use this DP seven as DP number one because it's a very, very important component that begins with preventive steps, and that's really what should be emphasized. #00:38:29#

Interviewer And now, the last one is bias mitigation and ethical decision-making. This principle is especially about ensuring fair treatment for all card users, so no group is unfairly treated. It's also about minimizing human bias in the decision-making process. In the context of decision-making, it's important to employ bias detection, perform continuous monitoring, and conduct regular audits and updates. This approach ensures that decisions are made without bias, fostering trust and reliability in the fraud detection system. For you as a compliance officer, it's crucial that the information you see isn't biased, so your decisions are always as accurate and fair as possible. What do you think about this principle? Does it make sense to you? Do you approve of it, or do you have any thoughts on how it could be improved? #00:39:08#

Interviewee Yeah, exactly. We are the decision-makers at the end of the process. We have the final say, yes or no, on whether something is a problem or fraud. It's always about making the superior decision, and that's just how it is. All the other tools and systems are just artificial aids to help us manage and monitor more effectively. Bias is really important here. Sometimes, the machine can actually be better at detecting certain things, but other times, a person might notice something from a different angle. However, as of now, human expertise in fraud detection is still the best approach. #00:40:18#

Interviewer Perfect. So, how do you feel about these design principles overall? Do you think they cover all the critical aspects of an AI-based fraud detection system? Is there anything you think should be added or improved? What are your overall thoughts on these proposed design principles? #00:42:13#

Interviewee I don't feel there's anything missing. I've had only a short time to think about it more deeply, but the principles as they are cover the key aspects of fraud prevention. That's everything we can put on the table to make the system functional. So, I think you've covered everything. Maybe the only thing I'd suggest is not about scalability, but the numbering, reflecting the criticality. For example, if DP one is the most important, it should be at the top, and the others should follow in descending order of importance. I'd consider moving something like DP seven up in priority, but otherwise, no, I think it all makes sense and is correct. #00:42:36#

Interviewer Perfect. Thank you so much. #00:44:03#

User 2 at amnis

Date: July 24th, 2024

Time: 10:30

Place: Microsoft Teams

00:00:00 – 00:31:02

Interviewer As I told you, this time we're going to evaluate my design principles that I've derived from literature and also from user interviews with users like you. This is really about getting your feedback and your feeling about these design principles so I can improve them accordingly based on your input. I'd like to start with the first design principle, which is about adaptive learning and continuous improvement. This principle is a requirement for such systems and aims to help developers improve fraud detection accuracy and performance over time for compliance officers like you. In the context of evolving transactional behaviors and changing card user habits, it's important to employ real-time learning mechanisms, analyses historical data, and continuously integrate user feedback. For example, if you identify something as fraud, the system should recognize it, and if you identify something as not fraud, the system can be adjusted to auto-close similar cases in the future. This involves implementing constant updates and refinement processes to adapt dynamically to both emerging fraud patterns and normal variations in card user behaviors. The goal is to ensure that the fraud detection system remains accurately tuned to fraudulent activities and legitimate user trends, thereby enhancing the system's reliability in a rapidly changing environment. So, this is the first design principle. I'd really like to know from you, do you have any suggestions for improvement, or what do you think about this first design principle concerning adaptive learning and continuous improvement in such systems? #00:00:06#

Interviewee I 100% agree with everything that's said there. This is exactly what we were talking about last time. If the AI or the system is working based on those learning curves, that's really the goal. I particularly like the last part about enhancing the system's reliability in a rapidly changing environment, and that's what it's all about. The way cards are being used is changing, not just across the whole customer base but for each individual customer. For example, they might learn they can use their card for something new, like paying for monthly subscriptions for services like Adobe Premium or something similar. Even such a change in transactional activity would currently trigger an alert for us, and it makes sense. If someone has only been using their card in supermarkets and then suddenly starts making larger online transactions, there should definitely be some sort of trigger. But once the compliance officer updates the card usage, the system should definitely take that into consideration. Agreed, agreed. Perfectly written. #00:01:59#

Interviewer Let's move to the next one, the second design principle, which is about user-centric development and alignment. At its core, this principle is about ensuring that compliance management is involved from the beginning of the development process. It's formulated for developers to create a tailored fraud detection system that aligns with the specific business model. For example, we have different clients than a private bank might have, with different services and goals. Our focus might be more on small and medium-sized enterprises. For managers and compliance officers, it's important that the fraud detection system development process includes active involvement from end users like us and incorporates regular feedback loops, whether through meetings or other channels, to tailor the system to specific business needs. This approach ensures that the fraud detection system is effectively aligned with strategic objectives and optimizes the cost-benefit ratio for each end user by meeting their specific goals, such as reducing manual work, increasing efficiency or speed, enhancing risk management, and making optimal use of our available resources, whether that's budget, personnel, or other assets. What do you think about this design principle?

Interviewee Again, there's not much to add here. I agree with what's mentioned. Reducing manual work is important, definitely, but on the other hand, the efficiency of the system still needs to be maintained. By that, I mean if the system runs fully automatically, I don't think that's something you'd want either. So, it's about finding the correct balance, optimizing everything without going too far in one direction. You can't just say, increase efficiency and speed, without considering that it's more about balancing different goals to achieve the best outcome. As an operator in charge of development and resolving alerts triggered by the automated system, you want the system to handle as much as possible. But at the same time, it's not feasible or advisable to have the system resolve everything on its own. In the end, an operator needs to make changes as needed, with the system adapting to that behavior. Finding the balance between automation and human intervention is the most important part. There's always room for improvement. Right now, in our scenario, we're more operator-oriented, but there's definitely space for improvement. Balance is key. #00:05:23#

Interviewer It's more about finding the right balance, and in the end, it must be a balanced approach that optimally uses the available resources, right? It's about considering how much money we spend, how much we can afford to spend, and how much human resources we have to tackle those alerts. And in the end, the ultimate goal, which wasn't mentioned here but came up in a discussion, is about protecting the customer and the company. That's the primary target. How you achieve that, whether through speed, efficiency, or other factors, requires a balanced approach. #00:07:42#

Interviewee It's about maintaining resources. On one hand, a fully automated system is going to cost the company some money. On the other hand, the operator who's resolving those alerts and triggers is also a cost. So really, it's about finding, as we've mentioned multiple times, the right cost-benefit ratio and optimizing it. #00:08:24#

Interviewer Thank you so much. Now, onto the next point, which is comprehensive data integration and quality assurance. This one is more on the technical side, but still really important. The idea here is to enhance fraud detection accuracy for compliance in the context of data quality and integrity. It's important to employ a full understanding of the data, comprehensive preprocessing techniques, advanced feature engineering, secure data handling, and, most importantly, the integration and analysis of various contextual data points. So, it's not just about seeing the transaction itself and the information you get from it, but also about considering factors like the time and place of the transaction, geolocation, login behaviours, device ID or IP address if a new device is used, and so on. All these should be integrated because it ensures that the model is built on a robust and reliable data foundation, allowing it to accurately identify fraudulent activities. What do you think about this design principle? #00:08:46#

Interviewee The more data you have to work with, the more efficient and better the outcome will be. In a world driven by data, it's essential to have good information for us to work with. There's always going to be noise around this data, which might not be as important, but specifically with debit cards and payment cards, there's a very broad access to information. Every POS is monitored and has its own specifications, so the data provided to operators who are resolving the triggered alerts, in our case at HAWK, is crucial. I can't even stress enough how important the data is because that's basically what I'm presented with to make decisions. If the data isn't sufficient, I can't make an informed decision. So, definitely, it's like the collection of evidence in any other case, the better the data, the better the outcome. #00:10:18#

Interviewer Thank you so much. Now, onto the next one, Design Principle four, which is the hybrid detection approach. This principle is about enhancing detection efficiency and adaptability for compliance officers in the context of operational needs. It's important to employ a hybrid approach that integrates both AI and rule-based methods. This combination provides flexibility; for example, if you have restricted countries where you don't want transactions to occur, you can simply set a rule to block any transaction in those countries. The same goes for setting amount limits, such as restricting a debit card from being used for transactions over a certain amount. This combination of AI and rules provides flexibility in adjusting detection rules and offers precision in fraud detection by leveraging the strengths of both methodologies. It improves accuracy and reduces manual review workload, ensuring that the system remains effective, responsive to evolving fraud patterns, and adaptable to changing requirements. What do you think about this design principle, which emphasizes the need for a hybrid approach between AI and rule-based methods? #00:11:52#

Interviewee I mean, as we mentioned in previous points, it's again about finding the perfect balance between these two approaches. Both AI and rule-based methodologies are important. There are certain scenarios where AI is going to be more effective, and others where the rule-based approach is necessary. That's why I think it's so interesting to work in this area. As transactional activity in the world changes, you need to adapt. I don't think there's ever a point where you can say, everything is set up perfectly as it should be, because there's always something that can be adapted or improved to give the best possible result and to protect everyone from fraudulent activity. So again, I'd say the hybrid approach, both AI and rule-based methodology, is important, and one cannot exist without the other. #00:13:12#

Interviewer Thank you so much. Now, the next one is about scalability and flexibility. To maintain long-term usability and effectiveness for compliance officers, especially in the context of increasing transaction volumes and evolving fraud techniques, it's important that the system is scalable. This means having a scalable infrastructure, a flexible system design, and conducting regular performance assessments. This approach ensures that the fraud detection system can efficiently handle growth, adapt to new fraud methodologies, and remain effective as the amount of data increases with higher transaction volumes. What do you think about this design principle? #00:14:46#

Interviewee I agree. Scalability is a very commonly used word nowadays, and it's definitely what you want to aim for. The system should be adaptable to a customer base of 100 clients as well as a customer base of 100'000 clients. In the end, whether you have 100 people or 10'000 customers, percentage-wise, the amount of scenarios should be very similar, it's just that the volume will differ. For the system to be flexible and able to adapt to such situations, I'd say that's very important. #00:15:33#

Interviewer Perfect. The next one is especially important for you as a compliance officer, it's about dynamic and interactive explainability. This principle focuses on enabling dynamic exploration of data and interactive explainability features for compliance officers when using the fraud detection system. It's important to provide user-friendly, interpretable insights, clear reasoning paths, and confidence levels in the system's decisions. This enhances trust, allows users to understand and validate the system's outputs, and supports effective decision-making in handling fraud cases. So, this principle is really about how you interact with the system. Do you think this design principle incorporates that interaction correctly? Does it capture what's needed? #00:16:39#

Interviewee I think it does. I consider myself quite technically capable, and yet every week I come across situations where I realize, oh, we can do that! It's really important that both sides of the coin, one being the developers of the system and the other being the users, cooperate properly. The effectiveness isn't just based on human capabilities. It's not just about how experienced you are as a developer or a compliance operator, but also about the cooperation between these two sides. For example, at Hawk, I'm still discovering new features and possibilities that, had I known about them a month ago, could have significantly eased my workload and the process of resolving alerts. That's why I believe quality-of-life changes for such programs and setups are very important. Again, it's about finding the balance between what's important for the end user and what's feasible both cost-wise and development-wise for the team. So, I definitely agree that finding balance and making the user experience as friendly as possible is crucial. #00:17:35#

Interviewer The next is about integrated preventive and detective measures. This principle is focused on enabling a strategic shift towards combining both preventive and detective fraud management for management and compliance officers in the context of comprehensive fraud protection. It's important to employ algorithms that support both preventive and detective measures because this approach allows, in the long term, for the system to use the same well-trained model, ensuring more secure protection for both card users and the company. The idea is that the fraud detection system should not only rely on either a preventive or detective approach but should, in the long term, combine both approaches to secure the company and, of course, the card users, our customers. What do you think about this design principle? #00:19:50#

Interviewee I agree. In a perfect world, we would prevent every single fraud, but that's impossible. I would say the bad guys are always one step ahead of the good guys. Again, it's about finding the balance between preventing fraud from happening and detecting it when it does. I would say it's as important as all of these points, really, none of them are less or more important. As we're talking about these specific points, I wouldn't rate one above the other. I wouldn't say number seven is more important than number four. It's all about finding balance in everything, and I don't want to repeat myself, but yes, I agree. There's really nothing more I can add or anything I would discredit in this. It's true, I agree. #00:20:51#

Interviewer Regarding the order of these design principles, it's not about ranking them from most important to least important, they are all of equal significance. It's really about finding a balance among all these principles to create an optimal fraud detection system. Now, onto the last design principle, Principle eight, which is about bias mitigation and ethical decision-making. This principle focuses on ensuring fair treatment of all card users, making sure that no group is unfairly targeted or criminalized, and minimizing human bias for compliance officers during the decision-making process. It's important to employ algorithms that detect and measure bias, perform continuous monitoring, and conduct regular audits and updates. This ensures that decisions are made without undue bias, fostering trust and reliability in the fraud detection system. It's crucial that the information shown to you isn't biased by the system, so your decisions aren't biased either. What do you think about this design principle? #00:22:20#

Interviewee I think I understand the meaning of this principle, but I'm not sure about the applicability of a system that would be biased. Do you mean it would trigger more alerts for a specific group of customers based on certain factors? To me, the AI making decisions and triggering alerts should always be focused on the data that's relevant to the transaction or payment itself. It's not really about fair or unfair treatment, but rather about acknowledging that there's a higher probability of fraudulent activity within specific groups of customers. It's not about being biased towards these groups, it's about staying true to the detection process, the data, and past experiences. On one hand, it's definitely important to ensure the system treats every customer group equally, but on the other hand, as I say that, I realize it can't really be that way. That's why the risk-based approach was developed, because you can't apply the same behavior towards one risk category as you would to another. They're ranked as low, medium, or high risk for a reason, and that's why the risk-based approach exists in the first place. So, the concept of fairness doesn't quite fit here. The system needs to be fair and ensure fair treatment, but at the same time, it has to recognize the differences in risk levels among different customer groups. It's a tricky balance to strike, and that's why this one is challenging for me to fully agree with, because there are really two scales that need to be balanced here. #00:23:25#

Interviewer I mean, that's true. The system also can't be biased by wrong decisions made by compliance officers, right? There's also that aspect to consider. If the system gets the wrong feedback, let's say you have an incompetent employee who doesn't carefully review cases and wrongly classifies activities as fraudulent,

then the system could end up being trained poorly based on that bad input. That kind of bias also needs to be controlled. #00:26:38#

Interviewee I get what you mean. If there were a compliance officer who was biased, whether due to incompetence or even being a fraudster themselves, it could create serious problems. There have been cases in the past where compliance officers willingly overlooked or even helped fraudsters, right? So, if there's a clear fraud attempt and it's successful, and the compliance officer feeds the AI with false data by treating those clear frauds as standard, non-fraudulent behavior, how is the system supposed to react and adapt to that situation? I don't even know the answer to that. I've never thought of it that way, really. #00:27:31#

Interviewer There are really two sides to bias, right? One is bias introduced by the system itself, whether due to its programming or the data it's trained on. For example, in our case, we deal with small and medium-sized enterprises, which is different from dealing with private clients or natural persons who behave differently. So the system might become biased toward small and medium-sized enterprise compliance, and then there's a bias when it comes to private individuals. That's one part. The other part is the potential for bias introduced by human decisions, whether it's wrong feedback or even false feedback provided by humans. So, there's human bias to consider as well. I think I need to make this clearer, that there are two aspects to consider: human bias and AI bias. Both need to be addressed. #00:28:34#

Interviewee And that's why I guess monitoring, along with both internal and external audits, is so important for the proper setup and longevity of the system. #00:29:31#

Interviewer In general, what do you think? Do you feel that these principles cover all the critical aspects of an AI-based fraud detection system? How do you feel about these design principles overall? Are they covering everything you think is important, or do you have any suggestions for improvements? #00:29:45#

Interviewee I mean, as we said, all of these eight points are applicable, and each of them plays a specific role and is important for the proper setup of a fraud detection system. One isn't more important than the other. I would say these principles cover the majority of what's important when it comes to a fraud detection system. To be honest, from the top of my head, it's difficult to think of anything else that wasn't mentioned here. Nothing really comes to mind, so I'd say this pretty much covers everything that should be covered when it comes to a fraud detection system. #00:30:01#

Interviewer Thank you so much for your time. #00:30:59#

User 3 at amnis

Date: Auguste 15th, 2024

Time: 10:15

Place: Microsoft Teams

00:00:00 – 00:15:03

- Interviewer** As I said before, in this interview we are going to discuss my derived design principles for AI-based fraud detection systems, derived from the current research, literature, and from user and expert interviews. Let's start with the first design principle, DP1, which focuses on adaptive learning and continuous improvement. For developers to improve fraud detection accuracy and performance over time for compliance officers in the context of evolving transactional behaviors and changing card user habits, employ real-time learning mechanisms, analyses historical data, and continuously integrate user feedback. Implement constant updating and refinement processes to adapt dynamically to both emerging fraud patterns and normal variations in card user behaviors. This approach ensures the FDS remains accurately tuned to fraudulent activities and legitimate user trends, enhancing the system's reliability in a rapidly changing environment. How well do you think this design principle will ensure the FDS remains effective over time by continuously adapting to new fraud patterns and user behaviors? Do you have any suggestions for improving this principle? #00:00:04#
- Interviewee** I think the principle is strong. Yes, but I would suggest also considering the inclusion of losses incurred. Such as maintaining a loss database or conducting case analysis. Additionally, it's important to consider situations where fraud was prevented by the appropriate actions of people involved. #00:01:22#
- Interviewer** Okay, perfect, thank you. Moving on to DP2, which is about user-centric development and alignment. For developers to create a tailored FDS that aligns with the business model, goals, services, and customer base for managers and compliance officers in the context of the FDS development process, employ active involvement of end-users and regular feedback loops. Tailor the system to specific business needs because this ensures the FDS is effective, aligned with strategic objectives, and optimises the cost-benefit ratio for each end-user by meeting their specific goals, such as reducing manual work, increasing efficiency or speed, or enhancing risk management, making optimal use of the available resources. Do you think this principle sufficiently addresses the need for alignment with business goals and user needs? How could it be enhanced? What do you think about this design principle? #00:02:23#
- Interviewee** Yes, I certainly agree with the approach. However, in practice, it will be a challenge, heavily dependent on the specific boundary conditions within an organization. Do you know what I mean? It might be useful to employ tools such as a fraud risk assessment to ensure that the model accurately covers the need. #00:03:22#
- Interviewer** I see. Perfect, that's a good suggestion. Now let's look at DP3, comprehensive data integration and quality assurance. For developers to enhance fraud detection accuracy for compliance officers in the context of data quality and integrity, employ thorough data understanding, comprehensive preprocessing techniques, advanced feature engineering, secure data handling, and integration and analysis of various contextual data points, such as time, place, login behaviors, device ID, IP address. This ensures the model is built on a robust and reliable data foundation, accurately identifying fraudulent activities. What do you think? How effective do you believe this design principle will be in ensuring high data quality and accuracy in fraud detection? Are there any additional measures or practices you would recommend? #00:03:57#
- Interviewee** Yes, yes, I agree with the design principle. Connecting the dots will definitely be helpful in validating decisions and enhancing the model. #00:05:02#
- Interviewer** Okay, perfect. Excellent. Let's move on to the next DP, DP4, which discusses a hybrid detection approach. For developers to enhance detection efficiency and adaptability for compliance officers in the con-text of operational needs, employ a hybrid approach that integrates AI and rule-based methods. This combination provides flexibility in adjusting detection rules and precision in fraud detection, leveraging the strengths of both methodologies to improve accuracy and reduce manual re-view workload. This ensures the system remains effective, responsive to evolving fraud patterns, and allows for timely adjustments to changing requirements. Do you think this principle strikes the right balance between AI and rule-based methods for fraud detection? Are there specific aspects of this integration that could be improved to better minimize false positives and enhance precision? What do you think of that? #00:05:32#

Interviewee I agree with the principle, but there's a tension between saving resources and the accuracy of the results. This tension doesn't offer much leeway for reducing the workload, which can increase risk for the organization and lead to a weaker model if less confirmed data points are used, potentially compromising the strategy. If you know what I mean. #00:06:24#

Interviewer I see, okay. Thank you so much. That's an important consideration. Let's move to the next DP. DP5 covers scalability and flexibility. For developers to maintain long-term usability and effectiveness for compliance officers in the context of increasing transaction volumes and evolving fraud techniques, employ scalable infra-structure, modular system design, and regular performance assessments. This ensures the FDS can efficiently handle growth, adapt to new fraud methodologies, and remain effective. Do you think this principle is sufficient to ensure long-term usability and effectiveness? What additional considerations would you recommend? Any suggestions? #00:06:53#

Interviewee I agree with the principle as it stands. #00:07:36#

Interviewer Thank you, perfect. Now let's discuss DP6, which is about dynamic and interactive explainability. For developers to enable dynamic exploration of data and interactive explainability features for compliance officers in the context of using the FDS, provide user-friendly, interpretable insights, clear reasoning paths, and confidence levels in the system's decisions. This enhances trust, allows users to understand and validate the system's outputs, and supports effective decision-making in handling fraud cases. How useful do you find the concept of dynamic and interactive explainability features for understanding FDS decisions? What improvements or additional features would make this principle more effective? #00:08:01#

Interviewee Yes, agreed. I strongly agree with this principle. However, it might be necessary to consider regulatory requirements needed to prove the effectiveness of the FDS. #00:09:14#

Interviewer Perfect. Okay. Yes, great insight. Moving on to DP7. It is about integrated preventive and detective measures. For developers to enable a strategic shift to combining preventive and detective fraud management for managers and compliance officers in the context of comprehensive fraud protection, employ FDS algorithms that support both preventive and detective measures. This allows the system to use the same well-trained model, ensuring more secure protection for card users and the company. Does this principle adequately address the need for predictive capabilities? What enhancements would you suggest? #00:09:43#

Interviewee Again, totally agree, yes. It is essential to transfer learnings from detective alerts into the preventive environment to enhance the effectiveness of preventive protection. Ideally, this would help avoid the financial impact of fraud before it occurs. #00:10:33#

Interviewer Okay, thank you so much. Finally, let's discuss DP8, the last DP, which focuses on bias mitigation and ethical decision-making. For developers to ensure fair treatment of all card users and minimize human bias for compliance officers in the context of decision-making processes, employ algorithms that detect and measure bias, perform continuous monitoring, and conduct regular audits and updates. This ensures decisions are made without undue bias, fostering trust and reliability in the FDS. Do you believe this principle adequately addresses the need for fair treatment and minimized human bias? How could it be strengthened? #00:11:14#

Interviewee I agree with the principle as it is currently formulated. Nothing to add here from my view. Totally agree. #00:12:14#

Interviewer Perfect. Thank you for that. As a final question, how do you feel about the design principles overall? #00:13:01#

Interviewee The design principles are well formulated and seem to be mutually exclusive with no overlaps. Agree with all of them. #00:13:23#

Interviewer Do you think these principles cover all critical aspects of an AI-based FDS? Or is something missing? What do you think? Any final suggestions? #00:13:23#

Interviewee Yes, I believe they do. Nothing to add. #00:14:00#

Interviewer Excellent. Perfect. Thank you again for your valuable feedback. Your insights will be incredibly helpful in refining these design principles. #00:14:16#